

Quantum Battleships with EV/Zeno Probes

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Abstract

We build a two-player *Quantum Battleships* game that uses interaction-free (Elitzur–Vaidman) probes implemented via Quantum Zeno sequences in Qiskit. Players choose board size, ship count, and a scan strategy (rows/cols/binary). Each probe yields one of **Found** / **Empty** / **Explosion**, and we score with an **EV** metric that rewards interaction-free detections while penalizing explosions and turns. An **optimizer** sweeps strategy and Zeno depth N and provides heatmaps/graphs to guide play.

1 Introduction

Motivation. Elitzur–Vaidman interaction-free measurement and Quantum Zeno dynamics let us gain information while suppressing destructive events. We cast these ideas as a duel-symmetric game: design probing policies that maximize information while minimizing explosions and resource cost. **Contributions.** (i) Playable two-player *Quantum Battleships* with adjustable board and probe knobs; (ii) a Qiskit Zeno-style probe (mid-circuit measure+reset); (iii) a fast analytic evaluator to sweep strategy and Zeno depth N ; (iv) an optimizer and plots (EV vs. N , heatmaps) yielding budget-aware picks. **Core idea.** A probe targets a subset on a hidden grid. Repeating N weak interrogations (small rotation on a path qubit, which-path marking, ancilla reset) returns **Found**, **Empty**, or **Explosion**. Primary score is **EV**; tie-breakers use **Turns** and **Gates**. [1, 2, 6]

2 Theory

IFM & Zeno. In interferometers, a potential absorber disturbs interference so a dark-port click can reveal it without interaction (IFM). Many weak interrogations place us in the Quantum Zeno regime, suppressing explosive events while preserving information [1, 2, 3, 4].

Circuit picture. One path qubit q_P + ancilla q_B . For $k = 1, \dots, N$: (i) small $R_y(2\theta)$ on q_P , (ii) CNOT $q_P \rightarrow q_B$, (iii) measure+reset q_B . If any ancilla = 1 $\Rightarrow$ **Explosion**. Otherwise measure q_P : 0 $\Rightarrow$ **Found**, 1 $\Rightarrow$ **Empty** [6].

Zeno scaling. With $\theta = \pi/(2N)$,

$$\Pr(\text{no explosion}) \approx \exp\left(-\frac{\pi^2}{4N}\right), \quad p_{\text{explode}} \approx \frac{\pi^2}{4N}.$$

Analytic shortcut used in sweeps: if region contains a ship use $\{p_{\text{explode}}, p_{\text{found}}, p_{\text{empty}}\} \approx \{\pi^2/(4N), 1 - \pi^2/(4N), 0\}$; if empty use $\{0, 0, 1\}$.

3 Scoring & Win Conditions

3.1 Primary Metrics and Updates

We track four running counters per player:

$$\# \text{Found}, \quad \# \text{Explosions}, \quad \text{Probes}, \quad \text{Ships remaining}.$$

After any positive signal (Found or Explosion), the probed region is subdivided and appended to the strategy queue; after an Empty, the region is discarded.

3.2 Win Conditions

A duel proceeds in *rounds* (Player A turn, then Player B turn) up to a configured round limit R .

Immediate victory. If a player reduces the opponent’s *Ships remaining* to 0 at any time, that player wins immediately.

Timeout resolution (when the round limit is reached). If neither side is eliminated by the end of round R , determine the winner by the following strict order:

1. Higher **EV** wins.
2. If EV ties, lower **Turns** wins.
3. If still tied, lower **Gates** wins.
4. If still tied, fewer **Probes** wins.
5. If all above tie, declare a **Draw**.

3.3 Design Notes

- The EV clamp $\max(\# \text{Explosions}, 1)$ prevents division by zero while still rewarding interaction-free detections.
- Larger N lowers p_{explode} (safer) but increases **Gates**; strategies that localize positives quickly (e.g., *binary*) often improve EV without excessive cost.
- Tie-breakers align with resource realism: **Turns** (time) before **Gates** (compute), then **Probes** (action count).

4 Methods

4.1 Board, State, and Outcomes

We sample an $M \times M$ grid with S non-overlapping 1×1 ships uniformly at random. A *probe* targets a subset of cells; its outcome is **Found**, **Explosion**, or **Empty**. If Explosion occurs, we remove one random ship from the probed subset (if any) and add P to the attacker’s Turns.

4.2 Region-Selection Strategies

We implement three deterministic scan policies with *adaptive refinement*:

- **rows**: queue holds M row segments; on a positive signal (Found/Explosion), split the segment in half and append halves (FIFO).
- **cols**: identical to rows but with column segments.
- **binary**: start from the full rectangle; on a positive, split along the longer side and append sub-rectangles (group-testing style) [5].

Stopping. A duel proceeds in rounds (A then B); we stop early if either board has zero ships, else at a configured round limit.

4.3 Probe Primitive (Aer)

For $k = 1, \dots, N$: $R_y(2\theta)$ on q_P with $\theta = \pi/(2N)$, then CNOT $q_P \rightarrow q_B$, then measure & reset q_B ; finally measure q_P . We map bitstrings as: any ancilla = 1 $\Rightarrow$ Explosion; else final $q_P = 0 \Rightarrow$ Found; else $q_P = 1 \Rightarrow$ Empty. Frequencies estimate $p_{\text{explode}}, p_{\text{found}}, p_{\text{empty}}$ for the chosen N .

4.4 Fast Analytic Evaluator (for Sweeps)

For rapid exploration we replace shot-based estimates with the Zeno-motivated shortcut of Eq. (??). If a subset contains ≥ 1 ship we sample from $\{\pi^2/(4N), 1 - \pi^2/(4N), 0\}$; when empty we use $\{0, 0, 1\}$. This preserves the *ordering* of strategies and depths while running quickly; absolute rates can be calibrated with Aer/hardware [2, 3, 4].

4.5 Symmetric Duel and Optimizer

To compare (strategy, N) fairly, both players adopt the same pair, play up to the round limit across seeds, and we average

$$\{\overline{\text{EV}}, \overline{\text{Turns}}, \overline{\text{Gates}}, \overline{\text{Probes}}\}.$$

The optimizer sweeps strategies $\times$ depths, aggregates trials, optionally filters by a Gates/Turns budget, and ranks by EV (then by cost and Probes).

4.6 Optimizer Details & Tuning

Inputs. Board size M and density (or S), strategy set $\mathcal{M}=\{\text{rows, cols, binary}\}$, candidate depths $\mathcal{N}$ (e.g., $\{4, 8, 12, 16, 24, 32\}$), round limit R , penalty P , trials T (seeds), and an optional budget: $\text{Gates} \leq B_g$ or $\text{Turns} \leq B_t$.

Metrics (per pair (m, N)). We run symmetric duels and average over trials:

$$\overline{\text{EV}}, \overline{\text{Turns}}, \overline{\text{Gates}}, \overline{\text{Probes}}, \quad \text{EV/Gate} = \frac{\overline{\text{EV}}}{\max(\overline{\text{Gates}}, 1)}, \quad \text{EV/Turn} = \frac{\overline{\text{EV}}}{\max(\overline{\text{Turns}}, 1)}.$$

Primary objective and tie-breakers. Maximize $\overline{\text{EV}}$ subject to the chosen budget (if any). Ties break by lower constrained cost (Gates or Turns) and then by fewer Probes. If multiple pairs remain indistinguishable, prefer the smaller N (simpler circuits).

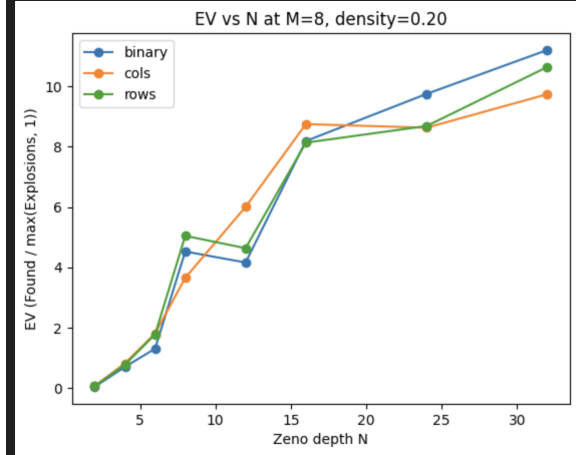
Ranking procedure.

1. For each $m \in \mathcal{M}$ and $N \in \mathcal{N}$, run T symmetric duels (round limit R) with distinct seeds.
2. Aggregate $\{\overline{\text{EV}}, \overline{\text{Turns}}, \overline{\text{Gates}}, \overline{\text{Probes}}\}$ and the derived ratios.
3. If a budget is set, filter rows to those with cost $\leq$ budget (fallback to unfiltered if empty).
4. Sort by $\overline{\text{EV}}$ (desc), then by the constrained cost (asc if budgeted), then by Probes (asc), then by N (asc).

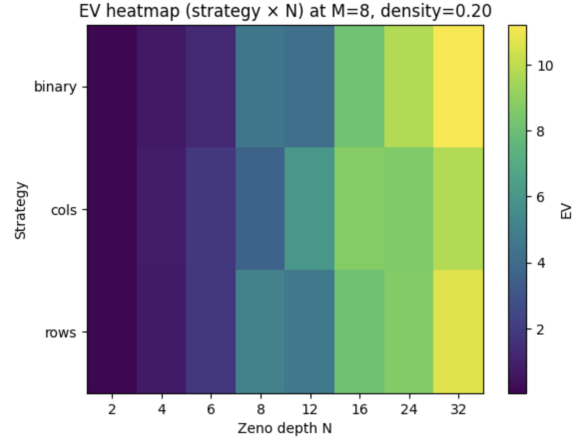
Defaults that work well . $M \in [8, 12]$; density 10–30%; $\mathcal{N}$ geometric (e.g., $\{4, 8, 12, 16, 24, 32\}$); $R \in [16, 24]$; $P \in \{1, 2, 3\}$; $T \in [6, 10]$. This usually identifies a *binary* strategy with moderate N as a strong baseline.

5 Results

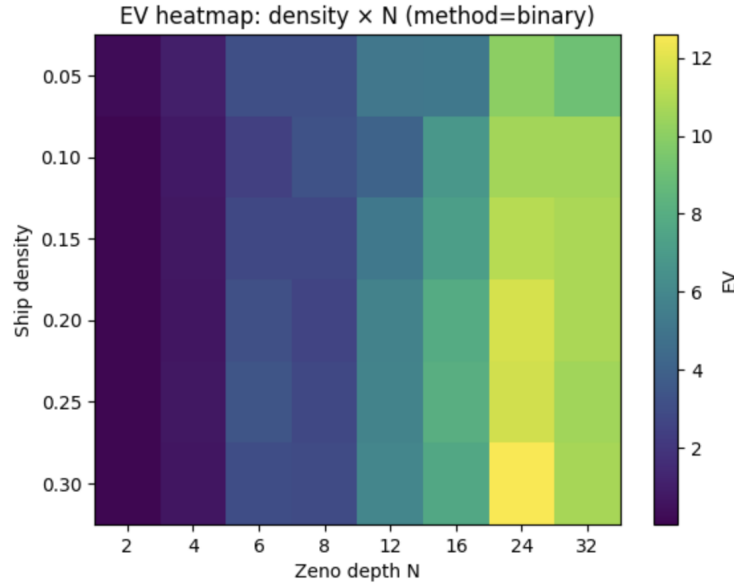
5.1 Figures



(a) **EV vs. N** by strategy. EV increases with N (fewer explosions), with diminishing returns.



(b) **Strategy $\times$ N heatmap.** EV of all 3 strategies compared with increasing Zeno depth.



(c) **Density $\times$ N (binary).** How ship density and N are related for binary method.

Figure 1: EV across strategies, Zeno depth N , and ship density.

5.2 Discussion

EV vs N . Fig. 1 (a) shows EV improving with N (fewer explosions) but with diminishing returns and rising Gates. (The increase in EV is less rewarding when $N > 15$).

Strategy comparison. Fig. 1 (b) shows a slight advantage for *column* at moderate N refinement. At very large N , *binary* has a slight advantage over the other 2 methods.

Cost-aware picks. If explosion penalty P is high (affecting **Turns**), favor larger N or binary. If gate-limited, use moderate/smaller N or apply a Gates budget; EV-per-gate often peaks mid-range.

6 Play & User Interface

6.1 Setup Flow (both players)

- **Board size M :** grid is $M \times M$.
- **Ships S or density:** number of hidden 1×1 ships (uniform without overlap).
- **Round limit:** maximum rounds (A then B) before timeout scoring (EV $\rightarrow$ Turns $\rightarrow$ Gates $\rightarrow$ Probes).
- **Explosion penalty P :** per-explosion addition to **Turns**.
- **Shots & seed (if simulating):** Aer shots for stability vs. runtime; seed for reproducibility.

6.2 Per-Player Controls (editable between turns)

- **Strategy** $\in \{\text{rows, cols, binary}\}$: rows/cols probe full lines and split on positives; binary probes rectangles and splits along the longer side (group testing).
- **Zeno depth N :** number of weak interrogations; larger N reduces p_{explode} but increases **Gates**.

6.3 In-Game Actions

- **Start:** initialize hidden boards and strategy queues.
- **Step A / Step B:** execute a single probe for the chosen player.
- **Play round:** run A then B (one full round).
- **Auto finish:** continue until a win or the round limit.
- **Reset / reseed:** new boards with current parameters (for batch experiments).

7 Conclusion

We demonstrated a duel-symmetric EV/Zeno game with tunable strategy and depth, plus an optimizer that exposes core trade-offs between safety (explosions), information (Found), and resource cost (Turns/Gates). The analytic evaluator gives rapid insight; for precision, swap in the Aer-based probe and noise models.

References

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